

SUFFOLK, VA, USA

WMO: 724007

Lat: 36.682N

Lon: 76.602W

Elev: 72

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03719

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	23.7	2.7	6.3	25.8	8.6	8.6	30.1	22.5	55.7	19.6	52.9	3.4	330	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	94.5	78.2	91.4	77.3	90.2	76.8	81.6	88.9	80.3	87.6	78.9	86.2	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	153.3	84.6	78.8	149.7	84.2	76.8	140.1	82.7	45.2	88.4	43.8	87.5	42.5	85.0	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.7	14.5	11.4	98.5	7.3	3.0	6.2	100.6	1.9	102.3	-2.2	104.0	-7.5	106.1
			10.5	83.3	6.7	2.6	5.7	85.2	1.8	86.7	-2.0	88.1	-6.8	90.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	40.6	43.8	50.3	60.2	68.5	76.3	79.4	78.4	72.9	62.4	52.0	45.3
	DBStd	15.82	11.05	9.56	9.85	8.67	7.34	5.58	4.59	4.32	5.57	8.24	8.91	9.44
	HDD50	984	335	220	123	12	0	0	0	0	0	6	79	209
	HDD65	3235	758	596	464	196	52	2	0	0	8	149	398	612
	CDD50	4981	43	47	132	319	575	789	912	882	687	391	140	64
	CDD65	1759	2	2	9	53	162	340	447	417	245	69	10	3
	CDH74	16261	5	16	90	508	1451	3292	4558	3907	1872	500	53	9
	CDH80	6326	0	0	12	141	484	1371	2008	1603	589	114	4	0
Wind	WSAvg	5.4	5.9	6.3	6.8	7.0	5.7	5.1	4.7	4.0	4.5	4.7	5.0	5.4
Precipitation	PrecAvg	50.0	3.7	3.0	4.0	3.3	4.0	4.3	5.5	5.4	5.5	3.9	3.5	3.5
	PrecMax	64.2	6.3	7.5	9.9	6.8	6.4	11.3	10.5	14.1	18.8	8.1	9.9	7.5
	PrecMin	33.2	1.1	1.1	0.6	0.9	1.9	1.6	2.1	1.5	0.5	0.0	0.4	1.4
	PrecStd	8.6	1.5	1.5	1.9	1.5	1.3	2.4	2.2	2.9	4.1	2.0	2.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.1	76.4	81.5	88.5	91.4	97.3	98.9	97.1	91.4	88.0	80.2	74.9
		MCWB	64.2	64.5	64.5	69.3	74.7	77.2	80.8	78.6	75.6	73.7	68.7	65.5
	2%	DB	69.6	71.0	75.5	83.8	88.3	93.0	94.8	93.0	88.7	82.8	74.8	70.2
		MCWB	63.5	60.6	62.2	67.6	72.8	77.2	79.0	78.5	75.0	71.3	65.5	65.2
	5%	DB	63.8	66.0	72.2	80.8	85.5	90.5	91.4	90.7	86.1	79.6	71.8	65.9
		MCWB	58.9	59.3	60.2	66.2	71.3	76.5	78.1	77.5	74.5	69.6	64.6	62.6
	10%	DB	59.0	61.7	67.7	75.2	82.0	88.1	90.1	88.3	83.5	75.4	67.3	62.7
		MCWB	53.6	54.1	59.6	64.0	70.0	75.2	77.6	76.8	73.2	67.9	62.1	58.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.4	66.8	68.8	73.1	77.8	83.4	83.3	83.1	80.4	77.2	72.4	69.4
		MCDB	71.8	72.5	75.2	80.3	87.8	89.9	93.0	91.1	83.3	82.4	75.1	72.4
	2%	WB	63.8	63.7	66.0	70.5	75.4	80.6	81.7	81.2	78.3	74.8	68.4	65.8
		MCDB	66.6	68.9	72.1	79.6	83.5	87.5	90.3	88.0	83.2	79.7	72.4	68.6
	5%	WB	60.1	60.4	63.5	68.5	73.7	78.4	80.4	79.8	77.0	72.8	65.8	63.0
		MCDB	63.8	65.3	68.4	76.7	81.1	86.6	88.7	86.7	82.8	77.3	70.2	65.0
	10%	WB	54.5	55.1	60.7	66.2	72.2	77.0	79.0	78.5	75.5	70.1	62.9	58.5
		MCDB	58.1	60.4	66.9	73.5	79.5	85.2	87.0	85.4	80.7	74.2	65.9	61.8
Mean Daily Temperature Range	5% DB	MDBR	18.8	20.6	21.0	22.3	20.4	19.1	18.2	17.7	18.0	20.5	21.0	19.0
		MCDBR	24.0	26.5	26.2	25.6	23.2	21.6	20.4	20.0	20.9	23.0	23.7	21.9
		MCWBR	18.4	18.4	15.1	11.8	10.2	8.6	8.3	7.9	9.6	12.1	15.2	17.1
	5% WB	MCDBR	22.0	25.0	22.0	21.9	19.9	18.8	18.3	17.3	16.5	17.7	19.3	19.9
		MCWBR	18.8	20.1	16.2	12.3	10.1	8.7	8.4	8.1	8.7	10.3	14.7	17.8
Clear-Sky Solar Irradiance	taub	0.306	0.320	0.347	0.390	0.435	0.488	0.514	0.507	0.405	0.356	0.323	0.310	
	taud	2.529	2.486	2.438	2.329	2.233	2.137	2.081	2.088	2.367	2.491	2.535	2.550	
	Ebn at Noon	283	292	292	283	271	255	247	246	269	274	274	274	
	Edh at Noon	25	29	34	40	44	49	51	50	36	29	25	23	
All-Sky Solar Radiation	RadAvg	763	997	1318	1682	1882	2014	1913	1724	1405	1128	856	678	
	RadStd	62	110	103	127	157	125	95	98	113	136	69	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	+301	+198

Nomenclature: See separate page